



3670 - Sinking silicates: tracing rainout across the LT transition

Cycle: 2, Proposal Category: GO

INVESTIGATORS

<i>Name</i>	<i>Institution</i>
Dr. Ben Burningham (PI) (ESA Member)	University of Hertfordshire
Dr. Jacqueline Kelly Faherty (CoI) (CoPI) (US Admin CoI)	American Museum of Natural History
Dr. Mark S. Marley (CoI)	University of Arizona
Dr. Channon Visscher (CoI)	Space Science Institute
Dr. John Michael Brewer (CoI)	San Francisco State University
Dr. Eileen Gonzales (CoI)	San Francisco State University
Dr. Johanna Vos (CoI) (ESA Member)	University of Dublin, Trinity College
Dr. Daniella Carolina Bardalez Gagliuffi (CoI)	Amherst College
Emily Calamari (CoI)	American Museum of Natural History
Mr. Austin James Rothermich (CoI)	American Museum of Natural History
Dr. Niall Whiteford (CoI)	American Museum of Natural History
Marina Gemma (CoI)	American Museum of Natural History
Caprice Lynette Phillips (CoI)	The Ohio State University
Prof. Kelle L. Cruz (CoI)	City University of New York Hunter College
Josefine Gaarn (CoI) (ESA Member)	University of Hertfordshire

OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
MIRI				
	1	HN PegB-MIRI	MIRI Low Resolution Spectroscopy	(1) V-HN-PEG-B
	6	HD46588-MIRI	MIRI Low Resolution Spectroscopy	(6) HD-46588-B
	7	HIP38939-MIRI	MIRI Low Resolution Spectroscopy	(7) HIP-38939-B
	18	HD89744B-MIRI	MIRI Low Resolution Spectroscopy	(8) HD89744B

JWST Proposal 3670 (Created: Saturday, February 1, 2025, 3:00:13AM Eastern Standard Time) - Overview

Folder	Observation	Label	Observing Template	Science Target
	19	HD16270B-MIRI	MIRI Low Resolution Spectroscopy	(9) HD16270B
	20	LSPMJ0632+5053B-MIRI	MIRI Low Resolution Spectroscopy	(10) LSPMJ0632+5053B
	21	NLTT1011B-MIRI	MIRI Low Resolution Spectroscopy	(11) NLTT1011B
	36	NLTT1011B-MIRI	MIRI Low Resolution Spectroscopy	(11) NLTT1011B
	22	GJ499C-MIRI	MIRI Low Resolution Spectroscopy	(12) GJ499C
	23	SDSS1416+50-MIRI	MIRI Low Resolution Spectroscopy	(13) SDSSJ141659.78+500626.4
	24	HD37216B-MIRI	MIRI Low Resolution Spectroscopy	(14) HD37216B
	25	HIP9269B-MIRI	MIRI Low Resolution Spectroscopy	(15) HIP9269B
	26	2106+3507-MIRI	MIRI Low Resolution Spectroscopy	(16) J210640.16+250729.0
NIRSPEC				
	11	HNPegB-NIRSPEC	NIRSpec Fixed Slit Spectroscopy	(1) V-HN-PEG-B
	16	HD46588-NIRSPEC	NIRSpec Fixed Slit Spectroscopy	(6) HD-46588-B
	17	HIP38939-NIRSPEC	NIRSpec Fixed Slit Spectroscopy	(7) HIP-38939-B
	27	HD89744B-NIRSPEC	NIRSpec Fixed Slit Spectroscopy	(8) HD89744B
	28	HD16270B-NIRSPEC	NIRSpec Fixed Slit Spectroscopy	(9) HD16270B
	29	LSPMJ0632+5053-NIRSPEC	NIRSpec Fixed Slit Spectroscopy	(10) LSPMJ0632+5053B
	30	NLTT1011B-NIRSPEC	NIRSpec Fixed Slit Spectroscopy	(11) NLTT1011B
	31	GJ499C-NIRSPEC	NIRSpec Fixed Slit Spectroscopy	(12) GJ499C
	32	SDSS1416+50-NIRSPEC	NIRSpec Fixed Slit Spectroscopy	(13) SDSSJ141659.78+500626.4
	33	HD37216B-NIRSPEC	NIRSpec Fixed Slit Spectroscopy	(14) HD37216B
	34	HIP9269B-NIRSPEC	NIRSpec Fixed Slit Spectroscopy	(15) HIP9269B
	35	2106+2507-NIRSPEC	NIRSpec Fixed Slit Spectroscopy	(16) J210640.16+250729.0

ABSTRACT

We will target a sample of abundance benchmarked brown dwarfs to create a transformative dataset that will impact substellar, exoplanetary and planetary science. JWST is unique in enabling us to:

- determine the dependence of cloud composition on Mg/Si, [M/H] and C/O to critically test cloud models

- track the rain out of oxygen across the $2000 > T_{\text{eff}} > 1000$ K range to empirically constrain calibrations for bulk C/O ratios from atmospheric observations of (exo)planets and brown dwarfs

Determinations of bulk compositions of giant planets are widely regarded as keystone evidence for understanding their formation and that of planetary systems. This work started in the Solar system and it continues to present challenges that limit the precision and accuracy of key diagnostics such as the C/O ratio. Relating measured atmospheric abundances to intrinsic (bulk) composition is not trivial. It requires a sound understanding of mixing, quenching and cloud condensation processes. Critically, this includes clouds that may be hidden from view but none-the-less alter the atmospheric composition.

This proposal will address these challenges by targeting a carefully selected sample of substellar benchmark binaries that span the same temperature range as directly imaged exoplanets. These systems will be used to calibrate the impacts of clouds, mixing and non-equilibrium chemistry on the measured atmospheric composition in the context of known bulk compositions determined from their stellar primaries. This project will create an empirical anchor for understanding how bulk compositions are imprinted on atmospheric abundances across a crucial regime of atmospheric physics.

OBSERVING DESCRIPTION

This project will obtain low-resolution NIRSpec Prism and MIRI LRS observations covering the full 0.6 - 12micron region for a sample of benchmark brown dwarfs spanning the entire LT sequence. These observations will achieve $\text{SNR} \sim 100$ in the flux peaks between water bands in the JH regions of the NIRSpec data, and $\text{SNR} \sim 50 - 100$ in the 8-9 micron region of the MIRI-LRS data.

We will use NIRPSpec in Prism mode, and will use the same detector setup for all targets, only changing the exposure times. We will use the WATA acquisition mode with the CLEAR filter and NRSRAPID readout pattern for target acquisition for brighter targets, or NRSRAPID6 for some fainter targets. We have ensured that we will obtain $\text{SNR} > 30$ for all targets in the NIRSpec target acquisition. For our science observations, we use the Prism/CLEAR grating/filter pair, the S200A1 slit (since each object is a point source) and will again use the NRSRAPID readout pattern. We will observe with an AB nod. We have used the JWST ETC to estimate exposure times for each target to achieve our required $\text{SNR} > \sim 100$ in the flux peaks.

For each target we used an ATMO2020 model of an appropriate T_{eff} and $\log g$ and Solar metallicity, scaled to match the target's photometry. We will observe in an AB nod pattern, and increase the Groups per integration until we achieve $\text{SNR} > 100$ in the flux peaks. Our longest exposure time for our faintest target is ~ 190 s.

We will use MIRI LRS in slit mode using the default P750L disperser and FULL subarray, as necessary for fixed-slit observations. We will use the FAST readout pattern, which is suitable for our purposes by ensuring no saturation while also providing adequate samples up the ramp. We will obtain an AB nod pattern. For target acquisition, we use the F1000W filter, the FULL subarray, FAST readout pattern and 4 groups for all targets. We will achieve $\text{SNR} > 40$ in acquisition for all targets.

For LRS observations we use the FULL subarray, FAST readout pattern and an AB nod pattern for sky subtraction. We have used the JWST ETC to estimate exposure times for each target to achieve the required $\text{SNR} > \sim 100$ in the 8 micron region, and used the same ATMO models as for the NIRSpec exposure estimates. Many of our targets necessarily have bright stars nearby, and we have taken care to specify orientation ranges to ensure that these bright sources do not fall in the MIRI imager during the LRS observation.

Proposal 3670 - Targets - Sinking silicates: tracing rainout across the LT transition

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	V-HN-PEG-B	RA: 21 44 28.7699 (326.1198746d) Dec: +14 46 6.02 (14.76834d) Equinox: J2000	Proper Motion RA: 231.108 mas/yr Proper Motion Dec: -113.2 mas/yr Parallax: 0.0551631" Epoch of Position: 2015.5712	
	Comments: Newly created target from Austin Category=Star Description=[Brown dwarfs] Extended=NO				
	(6)	HD-46588-B	RA: 06 46 26.9267 (101.6121946d) Dec: +79 34 54.54 (79.58182d) Equinox: J2000	Proper Motion RA: -99.015 mas/yr Proper Motion Dec: -603.924 mas/yr Parallax: 0.0549292" Epoch of Position: 2015.4041	
	Comments: Category=Star Description=[Brown dwarfs] Extended=NO				
	(7)	HIP-38939-B	RA: 07 58 1.7543 (119.5073096d) Dec: -25 39 3.13 (-25.65087d) Equinox: J2000	Proper Motion RA: 362.409 mas/yr Proper Motion Dec: -245.786 mas/yr Parallax: 0.0541012" Epoch of Position: 2015.4781	
	Comments: Category=Star Description=[Brown dwarfs] Extended=NO				
	(8)	HD89744B	RA: 10 22 14.7030 (155.5612625d) Dec: +41 14 24.27 (41.24008d) Equinox: J2000	Proper Motion RA: -119.215 mas/yr Proper Motion Dec: -140.39 mas/yr Parallax: .0258537" Epoch of Position: 2016	
	Comments: Category=Star Description=[Brown dwarfs] Extended=NO				
	(9)	HD16270B	RA: 02 36 0.0495 (39.0002062d) Dec: -23 31 20.06 (-23.52224d) Equinox: J2000	Proper Motion RA: 94.626 mas/yr Proper Motion Dec: 21.71 mas/yr Parallax: .046418" Epoch of Position: 2016	
	Comments: Category=Star Description=[Brown dwarfs] Extended=NO				
	(10)	LSPMJ0632+5053B	RA: 06 32 48.5892 (98.2024550d) Dec: +50 53 32.70 (50.89242d) Equinox: J2000	Proper Motion RA: 48.5 mas/yr Proper Motion Dec: -143.8 mas/yr Parallax: .0121095" Epoch of Position: 2015.4425	
	Comments: Category=Star Description=[Brown dwarfs] Extended=NO				

Proposal 3670 - Targets - Sinking silicates: tracing rainout across the LT transition

(11)	NLTT1011B	RA: 00 19 32.6410 (4.8860042d) Dec: +40 18 54.30 (40.31508d) Equinox: J2000	Proper Motion RA: -79.607 mas/yr Proper Motion Dec: -193.067 mas/yr Parallax: .0181393" Epoch of Position: 2016
Comments: Category=Star Description=[Brown dwarfs] Extended=NO			
(12)	GJ499C	RA: 13 05 41.0053 (196.4208554d) Dec: +20 46 40.76 (20.77799d) Equinox: J2000	Proper Motion RA: -50.183 mas/yr Proper Motion Dec: 82.325 mas/yr Parallax: .0502373" Epoch of Position: 2016
Comments: Category=Star Description=[Brown dwarfs] Extended=NO			
(13)	SDSSJ141659.78+500626.4	RA: 14 16 59.3901 (214.2474588d) Dec: +50 06 28.88 (50.10802d) Equinox: J2000	Proper Motion RA: -298.85 mas/yr Proper Motion Dec: 183 mas/yr Parallax: .02215" Epoch of Position: 2015.3411
Comments: Category=Star Description=[Brown dwarfs] Extended=NO			
(14)	HD37216B	RA: 05 39 49.5111 (84.9562963d) Dec: +52 53 57.38 (52.89927d) Equinox: J2000	Proper Motion RA: -13.403 mas/yr Proper Motion Dec: -151.063 mas/yr Parallax: .0355463" Epoch of Position: 2016
Comments: Category=Star Description=[Brown dwarfs] Extended=NO			
(15)	HIP9269B	RA: 01 59 11.0362 (29.7959842d) Dec: +33 12 25.92 (33.20720d) Equinox: J2000	Proper Motion RA: 245.7 mas/yr Proper Motion Dec: -352.7 mas/yr Parallax: .0404424" Epoch of Position: 2015.0397
Comments: Category=Star Description=[Brown dwarfs] Extended=NO			
(16)	J210640.16+250729.0	RA: 21 06 40.1629 (316.6673454d) Dec: +25 07 28.97 (25.12471d) Equinox: J2000	Proper Motion RA: -21.892 mas/yr Proper Motion Dec: -162.527 mas/yr Parallax: .029313 " Epoch of Position: 2015.5548
Comments: Category=Star Description=[Brown dwarfs] Extended=NO			

Proposal 3670 - Targets - Sinking silicates: tracing rainout across the LT transition

(17)	NLTT1011	RA: 00 19 34.4939 (4.8937246d) Dec: +40 18 2.93 (40.30081d) Equinox: J2000	Proper Motion RA: -79.06 mas/yr Proper Motion Dec: -190.375 mas/yr Parallax: .0181393" Epoch of Position: 2016.0
Comments: Category=Star Description=[K dwarfs] Extended=NO			

Proposal 3670 - Observation 1 - Sinking silicates: tracing rainout across the LT transition

Observation	Proposal 3670, Observation 1: HNPegB-MIRI								Sat Feb 01 08:00:13 GMT 2025
	Diagnostic Status: Warning								
	Observing Template: MIRI Low Resolution Spectroscopy								
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.								
Fixed Targets	#	Name	Target Coordinates		Targ. Coord. Corrections		Miscellaneous		
	(1)	V-HN-PEG-B	RA: 21 44 28.7699 (326.1198746d)		Proper Motion RA: 231.108 mas/yr				
			Dec: +14 46 6.02 (14.76834d)		Proper Motion Dec: -113.2 mas/yr				
			Equinox: J2000		Parallax: 0.0551631"				
					Epoch of Position: 2015.5712				
	Comments: Newly created target from Austin								
	Category=Star								
	Description=[Brown dwarfs]								
	Extended=NO								
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	1 V-HN-PEG-B	F1000W	FAST	4	1	1	11.1	63738.15
Template	Subarray				Obtain Verification Image?				
	FULL				false				
Dithers	#	Dither Type	No. Spectral Steps		Spectral Step Offset		No. Spatial Steps		Spatial Step Offset
	1	ALONG SLIT NOD							
Spectral Elements	#	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Exposures/Dith	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID
	1	FASTR1	35	1	2	1	2	194.253	63737.13

Proposal 3670 - Observation 1 - Sinking silicates: tracing rainout across the LT transition

Special Requirements	Aperture PA Range 235 to 280 Degrees (V3 230.24203 to 275.24203) Sequence Observations 1, 11, Non-interruptible
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Proposal 3670 - Observation 6 - Sinking silicates: tracing rainout across the LT transition

Observation	Proposal 3670, Observation 6: HD46588-MIRI								Sat Feb 01 08:00:13 GMT 2025
	Diagnostic Status: Warning								
	Observing Template: MIRI Low Resolution Spectroscopy								
Diagnostics	(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.								
Fixed Targets	#	Name	Target Coordinates		Targ. Coord. Corrections		Miscellaneous		
	(6)	HD-46588-B	RA: 06 46 26.9267 (101.6121946d) Dec: +79 34 54.54 (79.58182d) Equinox: J2000		Proper Motion RA: -99.015 mas/yr Proper Motion Dec: -603.924 mas/yr Parallax: 0.0549292" Epoch of Position: 2015.4041				
	Comments: Category=Star Description=[Brown dwarfs] Extended=NO								
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	6 HD-46588-B	F1000W	FAST	4	1	1	11.1	63738.15
Template	Subarray				Obtain Verification Image?				
	FULL				false				
Dithers	#	Dither Type	No. Spectral Steps		Spectral Step Offset		No. Spatial Steps		Spatial Step Offset
	1	ALONG SLIT NOD							
Spectral Elements	#	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Exposures/Dith	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID
	1	FASTR1	35	1	2	1	2	194.253	

Proposal 3670 - Observation 6 - Sinking silicates: tracing rainout across the LT transition

Special Requirements	Aperture PA Range 85 to 140 Degrees (V3 80.24203 to 135.24203) Sequence Observations 6, 16, Non-interruptible
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Proposal 3670 - Observation 7 - Sinking silicates: tracing rainout across the LT transition

Observation	Proposal 3670, Observation 7: HIP38939-MIRI								Sat Feb 01 08:00:13 GMT 2025
	Diagnostic Status: Warning								
	Observing Template: MIRI Low Resolution Spectroscopy								
Diagnostics	(Visit 7:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.								
Fixed Targets	#	Name	Target Coordinates		Targ. Coord. Corrections		Miscellaneous		
	(7)	HIP-38939-B	RA: 07 58 1.7543 (119.5073096d) Dec: -25 39 3.13 (-25.65087d) Equinox: J2000		Proper Motion RA: 362.409 mas/yr Proper Motion Dec: -245.786 mas/yr Parallax: 0.0541012" Epoch of Position: 2015.4781				
	Comments: Category=Star Description=[Brown dwarfs] Extended=NO								
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	7 HIP-38939-B	F1000W	FAST	4	1	1	11.1	63738.15
Template	Subarray				Obtain Verification Image?				
	FULL				false				
Dithers	#	Dither Type	No. Spectral Steps		Spectral Step Offset		No. Spatial Steps		Spatial Step Offset
	1	ALONG SLIT NOD							
Spectral Elements	#	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Exposures/Dith	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID
	1	FASTR1	35	1	2	1	2	194.253	

Proposal 3670 - Observation 7 - Sinking silicates: tracing rainout across the LT transition

Special Requirements	Aperture PA Range 90 to 50 Degrees (V3 85.24203 to 45.24203) Sequence Observations 7, 17, Non-interruptible
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Proposal 3670 - Observation 18 - Sinking silicates: tracing rainout across the LT transition

Observation	Proposal 3670, Observation 18: HD89744B-MIRI								Sat Feb 01 08:00:13 GMT 2025
	Diagnostic Status: Warning								
Diagnostics	Observing Template: MIRI Low Resolution Spectroscopy								
	(Visit 18:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.								
Fixed Targets	#	Name	Target Coordinates		Targ. Coord. Corrections		Miscellaneous		
	(8)	HD89744B	RA: 10 22 14.7030 (155.5612625d) Dec: +41 14 24.27 (41.24008d) Equinox: J2000		Proper Motion RA: -119.215 mas/yr Proper Motion Dec: -140.39 mas/yr Parallax: .0258537" Epoch of Position: 2016				
Acquisition	Comments: Category=Star Description=[Brown dwarfs] Extended=NO								
	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
Template	1	8 HD89744B	F1000W	FAST	4	1	1	11.1	63738.15
	Subarray				Obtain Verification Image?				
Dithers	FULL				false				
	#	Dither Type	No. Spectral Steps	Spectral Step Offset	No. Spatial Steps	Spatial Step Offset			
Spectral Elements	1	ALONG SLIT NOD							
	#	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Exposures/Dith	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID
	1	FASTR1	100	1	2	1	2	555.008	

Proposal 3670 - Observation 18 - Sinking silicates: tracing rainout across the LT transition

Special Requirements	Aperture PA Range 90 to 50 Degrees (V3 85.24203 to 45.24203) Sequence Observations 18, 27, Non-interruptible
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Proposal 3670 - Observation 19 - Sinking silicates: tracing rainout across the LT transition

Observation	Proposal 3670, Observation 19: HD16270B-MIRI								Sat Feb 01 08:00:13 GMT 2025
	Diagnostic Status: Warning								
Observing Template: MIRI Low Resolution Spectroscopy									
Diagnostics	(Visit 19:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.								
Fixed Targets	#	Name	Target Coordinates		Targ. Coord. Corrections		Miscellaneous		
	(9)	HD16270B	RA: 02 36 0.0495 (39.0002062d)		Proper Motion RA: 94.626 mas/yr				
			Dec: -23 31 20.06 (-23.52224d)		Proper Motion Dec: 21.71 mas/yr				
			Equinox: J2000		Parallax: .046418"				
					Epoch of Position: 2016				
	Comments: Category=Star Description=[Brown dwarfs] Extended=NO								
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	9 HD16270B	F1000W	FAST	4	1	1	11.1	63738.15
Template	Subarray				Obtain Verification Image?				
	FULL				false				
Dithers	#	Dither Type	No. Spectral Steps		Spectral Step Offset		No. Spatial Steps		Spatial Step Offset
	1	ALONG SLIT NOD							
Spectral Elements	#	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Exposures/Dith	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID
	1	FASTR1	25	1	2	1	2	138.752	

Proposal 3670 - Observation 19 - Sinking silicates: tracing rainout across the LT transition

Special Requirements	Aperture PA Range 90 to 50 Degrees (V3 85.24203 to 45.24203) Sequence Observations 19, 28, Non-interruptible
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Proposal 3670 - Observation 20 - Sinking silicates: tracing rainout across the LT transition

Observation	Proposal 3670, Observation 20: LSPMJ0632+5053B-MIRI								Sat Feb 01 08:00:13 GMT 2025
	Diagnostic Status: Warning Observing Template: MIRI Low Resolution Spectroscopy								
Diagnostics	(Visit 20:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.								
Fixed Targets	#	Name	Target Coordinates		Targ. Coord. Corrections		Miscellaneous		
	(10)	LSPMJ0632+5053B	RA: 06 32 48.5892 (98.2024550d) Dec: +50 53 32.70 (50.89242d) Equinox: J2000		Proper Motion RA: 48.5 mas/yr Proper Motion Dec: -143.8 mas/yr Parallax: .0121095" Epoch of Position: 2015.4425				
	Comments: Category=Star Description=[Brown dwarfs] Extended=NO								
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	10 LSPMJ0632+5053B	F1000W	FAST	4	1	1	11.1	63738.15
Template	Subarray				Obtain Verification Image?				
	FULL				false				
Dithers	#	Dither Type	No. Spectral Steps		Spectral Step Offset		No. Spatial Steps		Spatial Step Offset
	1	ALONG SLIT NOD							
Spectral Elements	#	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Exposures/Dith	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID
	1	FASTR1	100	1	2	1	2	555.008	

Proposal 3670 - Observation 20 - Sinking silicates: tracing rainout across the LT transition

Special Requirements	Aperture PA Range 90 to 50 Degrees (V3 85.24203 to 45.24203) Sequence Observations 20, 29, Non-interruptible
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Proposal 3670 - Observation 21 - Sinking silicates: tracing rainout across the LT transition

Observation	Proposal 3670, Observation 21: NLTT1011B-MIRI								Sat Feb 01 08:00:13 GMT 2025
	Diagnostic Status: Warning								
Observing Template: MIRI Low Resolution Spectroscopy									
Diagnostics	(Visit 21:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.								
Fixed Targets	#	Name	Target Coordinates		Targ. Coord. Corrections		Miscellaneous		
	(11)	NLTT1011B	RA: 00 19 32.6410 (4.8860042d) Dec: +40 18 54.30 (40.31508d) Equinox: J2000		Proper Motion RA: -79.607 mas/yr Proper Motion Dec: -193.067 mas/yr Parallax: .0181393" Epoch of Position: 2016				
Comments: Category=Star Description=[Brown dwarfs] Extended=NO									
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	11 NLTT1011B	F1000W	FAST	4	1	1	11.1	63738.15
Template	Subarray				Obtain Verification Image?				
	FULL				false				
Dithers	#	Dither Type		No. Spectral Steps	Spectral Step Offset		No. Spatial Steps	Spatial Step Offset	
	1	ALONG SLIT NOD							
Spectral Elements	#	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Exposures/Dith	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID
	1	FASTR1	100	1	2	1	2	555.008	

Proposal 3670 - Observation 21 - Sinking silicates: tracing rainout across the LT transition

Special Requirements	Aperture PA Range 90 to 50 Degrees (V3 85.24203 to 45.24203) Sequence Observations 21, 30, Non-interruptible
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Proposal 3670 - Observation 36 - Sinking silicates: tracing rainout across the LT transition

Observation	Proposal 3670, Observation 36: NLTT1011B-MIRI									Sat Feb 01 08:00:13 GMT 2025
	Diagnostic Status: Warning									
Observing Template: MIRI Low Resolution Spectroscopy										
Diagnostics	(Visit 36:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
	(Visit 36:1) Informational (Form): Visit schedulable, but most scheduling windows are when JWST is pointed in direction of greatest micrometeoroid impact risk. This is likely due to scheduling special requirements.									
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections		Miscellaneous		
	(11)	NLTT1011B	RA: 00 19 32.6410 (4.8860042d)			Proper Motion RA: -79.607 mas/yr				
			Dec: +40 18 54.30 (40.31508d)			Proper Motion Dec: -193.067 mas/yr				
			Equinox: J2000			Parallax: .0181393"				
						Epoch of Position: 2016				
	Comments: Category=Star Description=[Brown dwarfs] Extended=NO									
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	17 NLTT1011	F1500W	FAST	4	1	1	11.1	63738.15	
Template	Subarray					Obtain Verification Image?				
	FULL					true				
Dithers	#	Dither Type		No. Spectral Steps		Spectral Step Offset		No. Spatial Steps		Spatial Step Offset
	1	ALONG SLIT NOD								
Pointing Verification	#	PV Readout Pattern	PV Groups/Int	PV Integrations/Exp	PV Total Integrations	PV Exposures/Dith	PV Total Dithers	PV Total Exposure Time	PV ETC Wkbk.Calc ID	Filter
	1	FASTR1	5	1	1	1	1	13.875		F1000W

Proposal 3670 - Observation 36 - Sinking silicates: tracing rainout across the LT transition

Spectral Elements	#	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Exposures/Dith	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID
	1	FASTR1	100	1	2	1	2	555.008	
Special Requirements	Aperture PA Range 90 to 50 Degrees (V3 85.24203 to 45.24203)								

Proposal 3670 - Observation 22 - Sinking silicates: tracing rainout across the LT transition

Observation	Proposal 3670, Observation 22: GJ499C-MIRI								Sat Feb 01 08:00:13 GMT 2025	
	Diagnostic Status: Warning									
	Observing Template: MIRI Low Resolution Spectroscopy									
Diagnostics	(Visit 22:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
Fixed Targets	#	Name	Target Coordinates		Targ. Coord. Corrections		Miscellaneous			
	(12)	GJ499C	RA: 13 05 41.0053 (196.4208554d)		Proper Motion RA: -50.183 mas/yr					
			Dec: +20 46 40.76 (20.77799d)		Proper Motion Dec: 82.325 mas/yr					
			Equinox: J2000		Parallax: .0502373"					
					Epoch of Position: 2016					
	Comments: Category=Star Description=[Brown dwarfs] Extended=NO									
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	12 GJ499C	F1000W	FAST	4	1	1	11.1	63738.15	
Template	Subarray				Obtain Verification Image?					
	FULL				false					
Dithers	#	Dither Type		No. Spectral Steps		Spectral Step Offset		No. Spatial Steps		Spatial Step Offset
	1	ALONG SLIT NOD								
Spectral Elements	#	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Exposures/Dith	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID	
	1	FASTR1	35	1	2	1	2	194.253		

Proposal 3670 - Observation 22 - Sinking silicates: tracing rainout across the LT transition

Special Requirements	Aperture PA Range 90 to 50 Degrees (V3 85.24203 to 45.24203) Sequence Observations 22, 31, Non-interruptible
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Proposal 3670 - Observation 23 - Sinking silicates: tracing rainout across the LT transition

Observation	Proposal 3670, Observation 23: SDSS1416+50-MIRI								Sat Feb 01 08:00:13 GMT 2025
	Diagnostic Status: Warning								
	Observing Template: MIRI Low Resolution Spectroscopy								
Diagnostics	(Visit 23:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.								
Fixed Targets	#	Name	Target Coordinates		Targ. Coord. Corrections		Miscellaneous		
	(13)	SDSSJ141659.78+500626.4	RA: 14 16 59.3901 (214.2474588d) Dec: +50 06 28.88 (50.10802d) Equinox: J2000		Proper Motion RA: -298.85 mas/yr Proper Motion Dec: 183 mas/yr Parallax: .02215" Epoch of Position: 2015.3411				
	Comments: Category=Star Description=[Brown dwarfs] Extended=NO								
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	13 SDSSJ141659.78+500 626.4	F1000W	FAST	4	1	1	11.1	63738.15
Template	Subarray				Obtain Verification Image?				
	FULL				false				
Dithers	#	Dither Type	No. Spectral Steps		Spectral Step Offset		No. Spatial Steps		Spatial Step Offset
	1	ALONG SLIT NOD							
Spectral Elements	#	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Exposures/Dith	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID
	1	FASTR1	100	1	2	1	2	555.008	

Proposal 3670 - Observation 23 - Sinking silicates: tracing rainout across the LT transition

Special Requirements	Aperture PA Range 90 to 50 Degrees (V3 85.24203 to 45.24203) Sequence Observations 23, 32, Non-interruptible
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Proposal 3670 - Observation 24 - Sinking silicates: tracing rainout across the LT transition

Observation	Proposal 3670, Observation 24: HD37216B-MIRI								Sat Feb 01 08:00:13 GMT 2025
	Diagnostic Status: Warning								
Diagnostics	Observing Template: MIRI Low Resolution Spectroscopy								
	(Visit 24:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.								
Fixed Targets	#	Name	Target Coordinates		Targ. Coord. Corrections		Miscellaneous		
	(14)	HD37216B	RA: 05 39 49.5111 (84.9562963d) Dec: +52 53 57.38 (52.89927d) Equinox: J2000		Proper Motion RA: -13.403 mas/yr Proper Motion Dec: -151.063 mas/yr Parallax: .0355463" Epoch of Position: 2016				
Acquisition	Comments: Category=Star Description=[Brown dwarfs] Extended=NO								
	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
Template	1	14 HD37216B	F1000W	FAST	4	1	1	11.1	63738.15
	Subarray				Obtain Verification Image?				
Dithers	FULL				false				
	#	Dither Type	No. Spectral Steps	Spectral Step Offset	No. Spatial Steps	Spatial Step Offset			
Spectral Elements	1	ALONG SLIT NOD							
	#	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Exposures/Dith	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID
	1	FASTR1	25	1	2	1	2	138.752	

Proposal 3670 - Observation 24 - Sinking silicates: tracing rainout across the LT transition

Special Requirements	Aperture PA Range 90 to 50 Degrees (V3 85.24203 to 45.24203) Sequence Observations 24, 33, Non-interruptible
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Proposal 3670 - Observation 25 - Sinking silicates: tracing rainout across the LT transition

Observation	Proposal 3670, Observation 25: HIP9269B-MIRI								Sat Feb 01 08:00:13 GMT 2025
	Diagnostic Status: Warning								
	Observing Template: MIRI Low Resolution Spectroscopy								
Diagnostics	(Visit 25:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.								
Fixed Targets	#	Name	Target Coordinates		Targ. Coord. Corrections		Miscellaneous		
	(15)	HIP9269B	RA: 01 59 11.0362 (29.7959842d) Dec: +33 12 25.92 (33.20720d) Equinox: J2000		Proper Motion RA: 245.7 mas/yr Proper Motion Dec: -352.7 mas/yr Parallax: .0404424" Epoch of Position: 2015.0397				
	Comments: Category=Star Description=[Brown dwarfs] Extended=NO								
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	15 HIP9269B	F1000W	FAST	4	1	1	11.1	63738.15
Template	Subarray				Obtain Verification Image?				
	FULL				false				
Dithers	#	Dither Type	No. Spectral Steps		Spectral Step Offset		No. Spatial Steps		Spatial Step Offset
	1	ALONG SLIT NOD							
Spectral Elements	#	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Exposures/Dith	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID
	1	FASTR1	40	1	2	1	2	222.003	

Proposal 3670 - Observation 25 - Sinking silicates: tracing rainout across the LT transition

Special Requirements	Aperture PA Range 90 to 50 Degrees (V3 85.24203 to 45.24203) Sequence Observations 25, 34, Non-interruptible
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Proposal 3670 - Observation 26 - Sinking silicates: tracing rainout across the LT transition

Observation	Proposal 3670, Observation 26: 2106+3507-MIRI									Sat Feb 01 08:00:13 GMT 2025
	Diagnostic Status: Warning									
	Observing Template: MIRI Low Resolution Spectroscopy									
Diagnostics	(Visit 26:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections		Miscellaneous		
	(16)	J210640.16+250729.0	RA: 21 06 40.1629 (316.6673454d) Dec: +25 07 28.97 (25.12471d) Equinox: J2000			Proper Motion RA: -21.892 mas/yr Proper Motion Dec: -162.527 mas/yr Parallax: .029313 " Epoch of Position: 2015.5548				
	Comments: Category=Star Description=[Brown dwarfs] Extended=NO									
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	16 J210640.16+250729.0	F1000W	FAST	4	1	1	11.1	63738.15	
Template	Subarray					Obtain Verification Image?				
	FULL					false				
Dithers	#	Dither Type		No. Spectral Steps		Spectral Step Offset		No. Spatial Steps		Spatial Step Offset
	1	ALONG SLIT NOD								
Spectral Elements	#	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Exposures/Dith	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID	
	1	FASTR1	100	1	2	1	2	555.008		

Proposal 3670 - Observation 26 - Sinking silicates: tracing rainout across the LT transition

Special Requirements	Aperture PA Range 90 to 50 Degrees (V3 85.24203 to 45.24203) Sequence Observations 26, 35, Non-interruptible
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Proposal 3670 - Observation 11 - Sinking silicates: tracing rainout across the LT transition

Observation	Proposal 3670, Observation 11: HNPegB-NIRSPEC										Sat Feb 01 08:00:13 GMT 2025	
	Diagnostic Status: Warning											
	Observing Template: NIRSpec Fixed Slit Spectroscopy											
Diagnostics	(Visit 11:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(1)	V-HN-PEG-B	RA: 21 44 28.7699 (326.1198746d)			Proper Motion RA: 231.108 mas/yr						
			Dec: +14 46 6.02 (14.76834d)			Proper Motion Dec: -113.2 mas/yr						
			Equinox: J2000			Parallax: 0.0551631"						
						Epoch of Position: 2015.5712						
	Comments: Newly created target from Austin											
	Category=Star											
	Description=[Brown dwarfs]											
	Extended=NO											
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	SAME	WATA	SUB32	CLEAR	NRSRAPIDD6	3	1	1	0.26	63702.12	
Template	HFF Readout Mode			Slit			Subarray					
	false			S200A1			SUBS200A1					
Dithers	#	Primary Dither Positions						Sub-Pixel Pattern				
	1	3						NONE				
Spectral Elements	#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	PRISM/CLEAR	S200A1	NRSRAPID	10	1	1	NONE	3	3	51.475	63736.15

Proposal 3670 - Observation 11 - Sinking silicates: tracing rainout across the LT transition

Special Requirements	Sequence Observations 1, 11, Non-interruptible
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Proposal 3670 - Observation 16 - Sinking silicates: tracing rainout across the LT transition

Observation	Proposal 3670, Observation 16: HD46588-NIRSPEC										Sat Feb 01 08:00:13 GMT 2025	
	Diagnostic Status: Warning											
	Observing Template: NIRSpec Fixed Slit Spectroscopy											
Diagnostics	(Visit 16:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(6)	HD-46588-B	RA: 06 46 26.9267 (101.6121946d)			Proper Motion RA: -99.015 mas/yr						
			Dec: +79 34 54.54 (79.58182d)			Proper Motion Dec: -603.924 mas/yr						
			Equinox: J2000			Parallax: 0.0549292"						
	Comments: Category=Star Description=[Brown dwarfs] Extended=NO											
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	SAME	WATA	SUB32	CLEAR	NRSRAPIDD6	3	1	1	0.26	63702.12	
Template	HFF Readout Mode			Slit			Subarray					
	false			S200A1			SUBS200A1					
Dithers	#	Primary Dither Positions						Sub-Pixel Pattern				
	1	3						NONE				
Spectral Elements	#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	PRISM/CLEAR	S200A1	NRSRAPID	6	1	1	NONE	3	3	32.779	

Special Requirements	Sequence Observations 6, 16, Non-interruptible
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Proposal 3670 - Observation 17 - Sinking silicates: tracing rainout across the LT transition

Observation	Proposal 3670, Observation 17: HIP38939-NIRSPEC										Sat Feb 01 08:00:13 GMT 2025	
	Diagnostic Status: Warning											
	Observing Template: NIRSpec Fixed Slit Spectroscopy											
Diagnostics	(Visit 17:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(7)	HIP-38939-B	RA: 07 58 1.7543 (119.5073096d)			Proper Motion RA: 362.409 mas/yr						
			Dec: -25 39 3.13 (-25.65087d)			Proper Motion Dec: -245.786 mas/yr						
			Equinox: J2000			Parallax: 0.0541012"						
	Comments: Category=Star Description=[Brown dwarfs] Extended=NO											
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	SAME	WATA	SUB32	CLEAR	NRSRAPIDD6	3	1	1	0.26	63702.12	
Template	HFF Readout Mode			Slit			Subarray					
	false			S200A1			SUBS200A1					
Dithers	#	Primary Dither Positions						Sub-Pixel Pattern				
	1	3						NONE				
Spectral Elements	#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	PRISM/CLEAR	S200A1	NRSRAPID	12	1	1	NONE	3	3	60.823	

Proposal 3670 - Observation 17 - Sinking silicates: tracing rainout across the LT transition

Special Requirements	Sequence Observations 7, 17, Non-interruptible
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Proposal 3670 - Observation 27 - Sinking silicates: tracing rainout across the LT transition

Observation	Proposal 3670, Observation 27: HD89744B-NIRSPEC										Sat Feb 01 08:00:13 GMT 2025	
	Diagnostic Status: Warning											
	Observing Template: NIRSpec Fixed Slit Spectroscopy											
Diagnostics	(Visit 27:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(8)	HD89744B	RA: 10 22 14.7030 (155.5612625d)			Proper Motion RA: -119.215 mas/yr						
			Dec: +41 14 24.27 (41.24008d)			Proper Motion Dec: -140.39 mas/yr						
			Equinox: J2000			Parallax: .0258537"						
						Epoch of Position: 2016						
	Comments: Category=Star Description=[Brown dwarfs] Extended=NO											
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	SAME	WATA	SUB32	CLEAR	NRSRAPID	3	1	1	0.08	63702.12	
Template	HFF Readout Mode			Slit			Subarray					
	false			S200A1			SUBS200A1					
Dithers	#	Primary Dither Positions						Sub-Pixel Pattern				
	1	3						NONE				
Spectral Elements	#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	PRISM/CLEAR	S200A1	NRSRAPID	4	1	1	NONE	3	3	23.431	

Proposal 3670 - Observation 27 - Sinking silicates: tracing rainout across the LT transition

Special Requirements	Sequence Observations 18, 27, Non-interruptible
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Proposal 3670 - Observation 28 - Sinking silicates: tracing rainout across the LT transition

Observation	Proposal 3670, Observation 28: HD16270B-NIRSPEC										Sat Feb 01 08:00:13 GMT 2025	
	Diagnostic Status: Warning											
	Observing Template: NIRSspec Fixed Slit Spectroscopy											
Diagnostics	(Visit 28:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(9)	HD16270B	RA: 02 36 0.0495 (39.0002062d)			Proper Motion RA: 94.626 mas/yr						
			Dec: -23 31 20.06 (-23.52224d)			Proper Motion Dec: 21.71 mas/yr						
			Equinox: J2000			Parallax: .046418"						
						Epoch of Position: 2016						
	Comments: Category=Star Description=[Brown dwarfs] Extended=NO											
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	SAME	WATA	SUB32	CLEAR	NRSRAPID	3	1	1	0.08	63702.12	
Template	HFF Readout Mode			Slit			Subarray					
	false			S200A1			SUBS200A1					
Dithers	#	Primary Dither Positions						Sub-Pixel Pattern				
	1	3						NONE				
Spectral Elements	#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	PRISM/CLEAR	S200A1	NRSRAPID	4	1	1	NONE	3	3	23.431	

Proposal 3670 - Observation 28 - Sinking silicates: tracing rainout across the LT transition

Special Requirements	Sequence Observations 19, 28, Non-interruptible
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Proposal 3670 - Observation 29 - Sinking silicates: tracing rainout across the LT transition

Observation	Proposal 3670, Observation 29: LSPMJ0632+5053-NIRSPEC										Sat Feb 01 08:00:13 GMT 2025	
	Diagnostic Status: Warning											
	Observing Template: NIRSpec Fixed Slit Spectroscopy											
Diagnostics	(Visit 29:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(10)	LSPMJ0632+5053B	RA: 06 32 48.5892 (98.2024550d) Dec: +50 53 32.70 (50.89242d) Equinox: J2000			Proper Motion RA: 48.5 mas/yr Proper Motion Dec: -143.8 mas/yr Parallax: .0121095" Epoch of Position: 2015.4425						
	Comments: Category=Star Description=[Brown dwarfs] Extended=NO											
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	SAME	WATA	SUB32	CLEAR	NRSRAPID	3	1	1	0.08	63702.12	
Template	HFF Readout Mode			Slit			Subarray					
	false			S200A1			SUBS200A1					
Dithers	#	Primary Dither Positions						Sub-Pixel Pattern				
	1	3						NONE				
Spectral Elements	#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	PRISM/CLEAR	S200A1	NRSRAPID	10	1	1	NONE	3	3	51.475	

Proposal 3670 - Observation 29 - Sinking silicates: tracing rainout across the LT transition

Special Requirements	Sequence Observations 20, 29, Non-interruptible
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Proposal 3670 - Observation 30 - Sinking silicates: tracing rainout across the LT transition

Observation	Proposal 3670, Observation 30: NLTT1011B-NIRSPEC										Sat Feb 01 08:00:13 GMT 2025	
	Diagnostic Status: Warning											
	Observing Template: NIRSpec Fixed Slit Spectroscopy											
Diagnostics	(Visit 30:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(11)	NLTT1011B	RA: 00 19 32.6410 (4.8860042d)			Proper Motion RA: -79.607 mas/yr						
			Dec: +40 18 54.30 (40.31508d)			Proper Motion Dec: -193.067 mas/yr						
			Equinox: J2000			Parallax: .0181393"						
						Epoch of Position: 2016						
	Comments: Category=Star Description=[Brown dwarfs] Extended=NO											
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	SAME	WATA	SUB32	CLEAR	NRSRAPID	3	1	1	0.08	63702.12	
Template	HFF Readout Mode			Slit			Subarray					
	false			S200A1			SUBS200A1					
Dithers	#	Primary Dither Positions						Sub-Pixel Pattern				
	1	3						NONE				
Spectral Elements	#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	PRISM/CLEAR	S200A1	NRSRAPID	10	1	1	NONE	3	3	51.475	

Proposal 3670 - Observation 30 - Sinking silicates: tracing rainout across the LT transition

Special Requirements	Sequence Observations 21, 30, Non-interruptible
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Proposal 3670 - Observation 31 - Sinking silicates: tracing rainout across the LT transition

Observation	Proposal 3670, Observation 31: GJ499C-NIRSPEC										Sat Feb 01 08:00:13 GMT 2025	
	Diagnostic Status: Warning											
	Observing Template: NIRSpec Fixed Slit Spectroscopy											
Diagnostics	(Visit 31:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(12)	GJ499C	RA: 13 05 41.0053 (196.4208554d)			Proper Motion RA: -50.183 mas/yr						
			Dec: +20 46 40.76 (20.77799d)			Proper Motion Dec: 82.325 mas/yr						
			Equinox: J2000			Parallax: .0502373"						
						Epoch of Position: 2016						
	Comments: Category=Star Description=[Brown dwarfs] Extended=NO											
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	SAME	WATA	SUB32	CLEAR	NRSRAPID	3	1	1	0.08	63702.12	
Template	HFF Readout Mode			Slit			Subarray					
	false			S200A1			SUBS200A1					
Dithers	#	Primary Dither Positions						Sub-Pixel Pattern				
	1	3						NONE				
Spectral Elements	#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	PRISM/CLEAR	S200A1	NRSRAPID	6	1	1	NONE	3	3	32.779	

Proposal 3670 - Observation 31 - Sinking silicates: tracing rainout across the LT transition

Special Requirements	Sequence Observations 22, 31, Non-interruptible
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Proposal 3670 - Observation 32 - Sinking silicates: tracing rainout across the LT transition

Observation	Proposal 3670, Observation 32: SDSS1416+50-NIRSPEC										Sat Feb 01 08:00:13 GMT 2025	
	Diagnostic Status: Warning											
	Observing Template: NIRSpec Fixed Slit Spectroscopy											
Diagnostics	(Visit 32:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(13)	SDSSJ141659.78+500626.4	RA: 14 16 59.3901 (214.2474588d) Dec: +50 06 28.88 (50.10802d) Equinox: J2000			Proper Motion RA: -298.85 mas/yr Proper Motion Dec: 183 mas/yr Parallax: .02215" Epoch of Position: 2015.3411						
	Comments: Category=Star Description=[Brown dwarfs] Extended=NO											
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	SAME	WATA	SUB32	CLEAR	NRSRAPID	3	1	1	0.08	63702.12	
Template	HFF Readout Mode			Slit			Subarray					
	false			S200A1			SUBS200A1					
Dithers	#	Primary Dither Positions						Sub-Pixel Pattern				
	1	3						NONE				
Spectral Elements	#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	PRISM/CLEAR	S200A1	NRSRAPID	10	1	1	NONE	3	3	51.475	

Proposal 3670 - Observation 32 - Sinking silicates: tracing rainout across the LT transition

Special Requirements	Sequence Observations 23, 32, Non-interruptible
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Proposal 3670 - Observation 33 - Sinking silicates: tracing rainout across the LT transition

Observation	Proposal 3670, Observation 33: HD37216B-NIRSPEC										Sat Feb 01 08:00:13 GMT 2025	
	Diagnostic Status: Warning											
Diagnostics	Observing Template: NIRSspec Fixed Slit Spectroscopy											
	(Visit 33:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(14)	HD37216B	RA: 05 39 49.5111 (84.9562963d) Dec: +52 53 57.38 (52.89927d) Equinox: J2000			Proper Motion RA: -13.403 mas/yr Proper Motion Dec: -151.063 mas/yr Parallax: .0355463" Epoch of Position: 2016						
Acquisition	Comments: Category=Star Description=[Brown dwarfs] Extended=NO											
	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
Template	1	SAME	WATA	SUB32	CLEAR	NRSRAPID	3	1	1	0.08	63702.12	
	HFF Readout Mode			Slit			Subarray					
Dithers	false			S200A1			SUBS200A1					
	#	Primary Dither Positions						Sub-Pixel Pattern				
Spectral Elements	1	3						NONE				
	#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	PRISM/CLEAR	S200A1	NRSRAPID	4	1	1	NONE	3	3	23.431	

Proposal 3670 - Observation 33 - Sinking silicates: tracing rainout across the LT transition

Special Requirements	Sequence Observations 24, 33, Non-interruptible
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Proposal 3670 - Observation 34 - Sinking silicates: tracing rainout across the LT transition

Observation	Proposal 3670, Observation 34: HIP9269B-NIRSPEC										Sat Feb 01 08:00:13 GMT 2025	
	Diagnostic Status: Warning											
	Observing Template: NIRSpec Fixed Slit Spectroscopy											
Diagnostics	(Visit 34:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(15)	HIP9269B	RA: 01 59 11.0362 (29.7959842d) Dec: +33 12 25.92 (33.20720d) Equinox: J2000			Proper Motion RA: 245.7 mas/yr Proper Motion Dec: -352.7 mas/yr Parallax: .0404424" Epoch of Position: 2015.0397						
	Comments: Category=Star Description=[Brown dwarfs] Extended=NO											
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	SAME	WATA	SUB32	CLEAR	NRSRAPID	3	1	1	0.08	63702.12	
Template	HFF Readout Mode			Slit			Subarray					
	false			S200A1			SUBS200A1					
Dithers	#	Primary Dither Positions						Sub-Pixel Pattern				
	1	3						NONE				
Spectral Elements	#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	PRISM/CLEAR	S200A1	NRSRAPID	10	1	1	NONE	3	3	51.475	

Proposal 3670 - Observation 34 - Sinking silicates: tracing rainout across the LT transition

Special Requirements	Sequence Observations 25, 34, Non-interruptible
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Proposal 3670 - Observation 35 - Sinking silicates: tracing rainout across the LT transition

Observation	Proposal 3670, Observation 35: 2106+2507-NIRSPEC										Sat Feb 01 08:00:13 GMT 2025	
	Diagnostic Status: Warning											
	Observing Template: NIRSspec Fixed Slit Spectroscopy											
Diagnostics	(Visit 35:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(16)	J210640.16+250729.0	RA: 21 06 40.1629 (316.6673454d) Dec: +25 07 28.97 (25.12471d) Equinox: J2000			Proper Motion RA: -21.892 mas/yr Proper Motion Dec: -162.527 mas/yr Parallax: .029313 " Epoch of Position: 2015.5548						
	Comments: Category=Star Description=[Brown dwarfs] Extended=NO											
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	SAME	WATA	SUB32	CLEAR	NRSRAPID	3	1	1	0.08	63702.12	
Template	HFF Readout Mode			Slit			Subarray					
	false			S200A1			SUBS200A1					
Dithers	#	Primary Dither Positions						Sub-Pixel Pattern				
	1	3						NONE				
Spectral Elements	#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	PRISM/CLEAR	S200A1	NRSRAPID	10	1	1	NONE	3	3	51.475	

Special Requirements	Sequence Observations 26, 35, Non-interruptible
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